Self.AI

Simulating Autonomous Vehicles

CARLA • LGSVL • AirSim • SUMO

Virtual Reality, Augmented Reality and Haptics

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Tasks

Python

Setup:

- Runing CARLA server
- 1 Ego vehicle
- 20 static NPCs around on different distance from the ego
- Lidar attached to the ego vehicle

<u>Visualize point cloud and Ground Truth 3D Bounding Boxes with Open3D</u>

LiDAR in CARLA — Key Parameters

- channels: number of laser beams (64— mirrors Velodyne model)
- range: max detection distance in metres (default 100 m)
- points_per_second: total point rate across all beams (e.g. 500,000)
- upper_fov / lower_fov: vertical field of view in degrees (e.g. 10 / -30)
- noise_stddev: Gaussian noise on returned distances (realistic sensor modeling)
- Semantic LiDAR also returns: object_tag, object_idx per point — free ground truth!

Optional: store dataset

- PCDs - .pcd files
- Labels - .json https://github.com/naurril/SUSTechPOINTS\

Store dataset in the Sustechpoints format and visualize